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Dual developmental effects of ARX poly-alanine mutations on human cortical excitatory and inhibitory neurons

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AUTHOR CONTRIBUTIONS

Conceptualization, V.N.-E., P.V., Z.R.L., D.M.T. and J.H.; methodology, V.N.-E., P.V., Z.R.L., and J.H.; software, P.V.; formal analysis, V.N.-E., P.V., S.M., A.H., and Z.R.L.; investigation, V.N.-E., P.V., S.M., Z.R.L., J.C., S.G.-A., C.N., M.J.S., and S.G.; resources, C.N. and D.M.T., writing – original draft, V.N.-E.; writing – review & editing, V.N.-E., P.V., S.M., and J.H.; visualization, V.N.-E., P.V., and J.H.; supervision, V.N.-E., M.G., and J.H.; funding acquisition, V.N.-E., P.V., and J.H.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Jenny Hsieh (jenny.hsieh@utsa.edu).

Materials availability

All unique/stable reagents generated in this study are available from the lead contact with a completed material transfer agreement.

Data and code availability

- The scRNA-seq datasets have been deposited at GEO and are available under accession number GEO: GSE215362.
- The codes used in this study can be found at https://github.com/parulvarma123/ARX_Organoids_scRNAseq, https://doi.org/10.5281/zenodo.17426624, and https://github.com/mgiugliano/SpiQ.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

DECLARATION OF INTERESTS

The authors declare no competing interests.

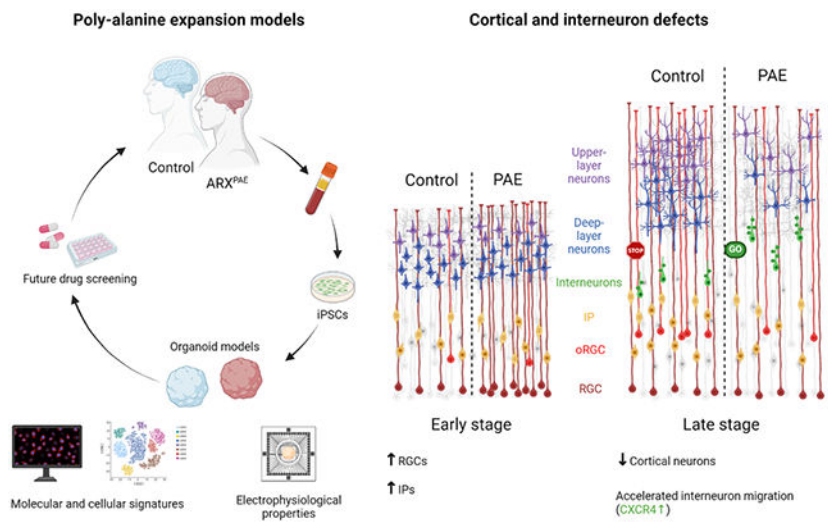
DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used Copilot in order to revise grammar and refine the text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility.

SUMMARY

Infantile spasms (IS), a severe childhood epilepsy with an incidence of 1.6–4.5 per 10,000 live births, often lead to lifelong intellectual disability. Up to 5% of affected males carry mutations in the Aristaless-related homeobox (*ARX*) gene. The lack of human-specific models for developmental epilepsy limits progress, making organoids a promising alternative. We use human cortical organoids (COs) and ganglionic eminence organoids (GEOs) to model poly-alanine expansion (PAE) mutations in *ARX*. PAE mutations increase cortical progenitor proliferation and accelerate early cortical development. *ARX* expression is upregulated in patient-derived COs at 30 days *in vitro* (DIV), correlating with altered cell cycle gene expression. We observe enhanced, cell-autonomous interneuron migration, which is rescued by CXCR4 inhibition. ARX^{PAE} assembloids exhibit early network hyperactivity. These findings highlight the utility of human brain organoids in uncovering ARX^{PAE}-driven mechanisms and represent a critical step toward developing targeted therapies for IS and related developmental epilepsies.

Graphical Abstract



In brief

Nieto-Estevez et al. show that ARX polyalanine expansion mutations disrupt cortical and interneuron development in human brain organoids, revealing early network hyperactivity and offering insights into developmental epilepsies.

INTRODUCTION

Infantile spasms are a severe childhood epilepsy affecting 1.6–4.5 per 10,000 live births, often causing lifelong intellectual disability (ID). About 5% of affected males carry mutations in the X-linked Aristaless-related homeobox (*ARX*) gene,[1,2] a transcription factor with a homeodomain (HD) and four poly-alanine (PA) tracts.[3-6] *ARX* mutations cause diverse phenotypes: HD or truncating mutations lead to severe ID, autism spectrum disorder (ASD), seizures, and brain malformation,[7-9] while missense and poly-alanine expansions (PAEs) in the first two tracts cause ID, ASD, and epilepsy without major structural defects.[8,10,11] PAEs have been identified in nine genes, mostly transcription factors, and are typically shorter than poly-glutamine repeats and lead to similar developmental disorders, suggesting shared mechanisms.[12,13]

During development, *Arx* is expressed in the nervous system, pancreas, and testes. In adult mice, it is found in the brain, muscle, heart, and liver.[14,15] In the brain, *Arx* is expressed in both cortical progenitor cells and postmitotic cortical γ-aminobutyric acid-containing (GABAergic) interneurons, where it plays critical roles during interneuron migration and cortical development.[14,16,17] Studies using knockout (KO) mouse models or mutations in the HD of *Arx* have recapitulated the phenotypes seen in patients with the most severe forms of ARX-associated epilepsy. These mice show aberrant tangential migration and a reduction of GABAergic interneurons in the cortex and other brain regions.[18,19] However, perinatal lethality has precluded analysis of adolescent or adult mice.[15] To overcome this limitation, conditional KO (cKO) mice lacking *Arx* specifically in interneuron progenitor cells have been generated; these mice show a similar loss of interneurons[20-22] and develop seizures.[20] Loss of *Arx* in cortical projection neuron progenitor cells results in a proliferation defect and a reduction in upper-layer cortical neurons.[16,17,23] Interestingly, mice lacking *Arx* in cortical projection neuron precursors do not develop seizures, although they exhibit hyperactivity and behavioral abnormalities.[24] In contrast, mice carrying a PAE mutation exhibit fewer cortical GABAergic interneurons[19,25] and develop seizures.[26-29] Although these mouse models have advanced our understanding of ARX function in neurological disorders, they fail to fully recapitulate human brain development, as mice lack key aspects of human cortical complexity. This limitation is particularly evident in drug development, where treatments effective in animal models often fail in clinical trials.[30] Moreover, epilepsy in many patients with *ARX* mutations remains resistant to available pharmaceutical therapies. Thus, there is a clear need for human-based models for studying neurological diseases and drug discovery.

Postmortem tissues from patients with rare neurological diseases, particularly pediatric patients, are rarely available for study. Neural organoids are excellent *in vitro* models for studying human development, as they recapitulate key features such as the presence of outer radial glial cells (oRGCs), expansion of cortical layers, and interneuron migration.[31-35] Importantly, organoid-based approaches also provide new opportunities to develop and test treatments for human neurological disorders.[36-38]

To examine ARX PAE mutations in human brain development and epilepsy, we recruited three male patients with a second-tract PAE and three healthy controls. Patient-derived human induced pluripotent stem cells (hiPSCs) were used to generate cortical organoids (COs) and ganglionic eminence organoids (GEOs) following established protocols.[32,35,39] We applied immunohistochemistry, single-cell RNA sequencing, and multi-electrode array recordings to assess cortical and interneuron development. Our results revealed that ARX^{PAE} mutations upregulate ARX in COs at 30 days *in vitro* (DIV), driving increased progenitor proliferation, reduced *CDKN1C* expression, and activation of cell cycle genes. Cortical progenitors also advance faster along their developmental trajectory. In GEOs, PAE reduces ARX^+ and GABAergic neuron numbers but accelerates interneuron migration, which can be rescued by CXCR4 inhibition. Multi-electrode array recordings reveal disrupted neuronal circuitry and heightened network activity, indicating early hyper-excitability. These results underscore ARX's dual role in human brain development and highlight organoids as a platform for mechanistic insights and therapy discovery.

RESULTS

Generation and characterization of COs and GEOs

ARX is a transcription factor essential for brain development, regulating cortical progenitor differentiation and interneuron migration.[14,16,17] While most insights come from mouse models, their relevance to human development remains uncertain. To address this, we used COs and GEOs as human-specific *in vitro* models. To confirm that our organoids accurately reproduced the developmental sequence observed by others, we performed immunohistochemistry (IHC) and single-cell RNA sequencing (scRNA-seq). Neural differentiation was induced through SMAD inhibition to generate COs from hiPSCs[35] (Figure 1A). scRNA-seq analysis confirmed the generation of two distinct types of organoids at 30 DIV and 120 DIV (Figures 1B and S1). At 30 DIV, COs contained the main cell types found in the developing cortex, including RGCs, cycling progenitor cells, intermediate progenitors (IPs), deep-layer neurons, a population of early Cajal-Retzius cells, and some interneurons (Figure 1C). Moreover, IHC analysis showed that COs at early time points (30 and 60 DIV) had ventricle-like structures, defined by PAX6$^+$ (Paired Box 6), SOX2$^+$ (SRY-box transcription factor 2), and VIMENTIN$^+$ cells (markers of dorsal cortical progenitor cells[40,41]), surrounded by maturing neurons labeled with CTIP2 and TBR1 (markers of deep-layer cortical neurons[42-44]) (Figures 1D, 1E, S2A, S3A, and S3B). Similar analysis of COs at 120 DIV revealed the presence of oRGCs marked by expression of *FAM107A* (family with sequence similarity 107 member A) and *HOPX*, key factors expressed in human oRGCs.[45,46] We also observed expression of the cortical neuron markers FEZF2, CTIP2, and TBR1 as well as BRN2 (*POU3F2*), SATB2, CUX1, CUX2, LHX5, and REELIN in COs at 120 DIV by IHC and/or scRNA-seq (Figures 1C, 1F, S2B, and S3C). While FEZF2, CTIP2, and TBR1 labeled deep-layer neurons, BRN2 (*POU3F2*), SATB2, CUX1, CUX2, LHX5, and REELIN were expressed in upper-layer neurons.[43,47-49] Moreover, the expression of *EMX1* and *vGLUT1*, but not *NKX2.1*, measured by RT-qPCR, as well as the presence of *SLC17A7* (vGLUT1) and *SLC176* (vGLUT2) observed by scRNA-seq, confirmed the dorsal identity of COs and the presence of glutamatergic neurons (Figures S2 and S3D). Consistent with the presence of interneurons, we also detected low expression of *vGAT*, *GAD1*, and *GAD2* in COs (Figures S2 and S3D). The distribution of rosettes varied between organoids; in some, rosettes were uniformly distributed, while in others, particularly larger organoids, they localized to specific regions by 30 DIV (Figure S3A). These rosettes remained visible at 60 DIV but became more diffuse at 120 DIV (Figures S3B and S3C). We also generated GEOs resembling the ganglionic eminence (GE) by modulating WNT (Wingless-related integration site) and SHH (Sonic Hedgehog) pathways through addition of IWP-2 and SAG (Smoothened receptor agonist).[32] We found that GEOs at 30 DIV predominantly expressed medial ganglionic eminence (MGE) markers (e.g., *NKX2.1*), while a smaller percentage of cells expressed caudal ganglionic eminence (CGE)/lateral ganglionic eminence (LGE) markers (e.g., *GSX2*) (Figures 1G, 1H, S4, and S5A).[50,51] Our analysis of GEOs at 120 DIV, a later developmental time point, revealed TUJ1$^+$, GABA$^+$, and CXCR4$^+$ neurons (Figures 1I and S5B). CXCR4 (C-X-C motif chemokine receptor 4) has been shown to play an important role in interneuron migration.[52-54] Moreover, the expression of *NKX2.1* and *vGAT*, but not *EMX1* or *vGLUT1*, confirmed the inhibitory nature of the neurons in the GEOs (Figure S5C). We also detected cells expressing calbindin (CB), calretinin (CR), and somatostatin (SST) (Figures 1J and S6), three markers of interneuron subtypes.[51] In addition, we found a small percentage of cells expressing gene signatures of microglia and oligodendrocyte progenitor cells at 30 and 120 DIV, respectively. Heatmaps showing the specific gene sets expressed by each cluster or cell type identified by scRNA-seq in COs and GEOs at 30 and 120 DIV are shown in Figure S7 and Data S1. These results corroborate the generation of neural organoids resembling cortical and subpallial regions of the human brain, as described previously.[32,35,55]

Expression of ARX and impact of PAE in COs and GEOs

We first interrogated our scRNA-seq dataset and found that a similar percentage of cells expressed *ARX* in control COs and GEOs at 30 and 120 DIV (Figure 2A; $p = 0.6593$). Furthermore, and consistent with known data, for the cells expressing *ARX* in COs, a high percentage were proliferative at the earlier time point (Figure 2B).[14] We also observed progenitor cells and cortical neurons expressing *ARX* in COs at both time points. In GEOs, *ARX* was expressed in both progenitor cells and GABAergic neurons (Figure 2C). We further confirmed the expression of ARX in progenitor cells (Ki67) and in neurons (doublecortin [DCX] and TUJ1) in COs and GEOs by IHC (Figures 2D and 2E). However, the lack of compatible antibodies prevented us from analyzing the co-localization of ARX with other specific markers (e.g., PAX6, SOX2, NKX2.1, and GABA). These qualitative data were consistent with the expression of *ARX* in human brain tissue (Figure S8). Thus, during embryonic development, *ARX* expression is higher in the GE/striatum than in the cortex, similar to data from mice.[14] Furthermore, *ARX*-expressing cells can be found both in the cortex and the striatum throughout adult life, although the level of expression is lower than in embryonic tissues.

To study the role of ARX during human development, we recruited three healthy controls and three patients with eight additional alanines in the second polyalanine tract of *ARX* (Table S1). We focused on male patients because *ARX* is located on the X chromosome, and females typically present with milder and more variable phenotypes.[20] Six lines of hiPSCs were generated (one per person) and two clones per line were used in all experiments (or as otherwise indicated). All lines expressed pluripotency markers (NANOG, SOX2, OCT4, LIN28, SSEA-4, and TRA-1-6), lost reprogramming factors (as shown by the lack of expression of the vectors used for reprograming), and maintained normal karyotype throughout the study (Figures S9A-S9D). The number of alanines was confirmed by PCR amplification followed by Sanger sequencing (Figure S9E). Then, we first assayed ARX expression in COs and GEOs derived from patients (Figures 2F-2I). We found a significant increase in ARX expression among cells in ARX^{PAE} COs at 30 DIV when compared to controls ($p = 0.0105$, d = 0.944; Figures 2F and 2G). scRNA-seq analysis confirmed this increase in *ARX* expression in COs in all cell types (Figure S10), and this finding was consistent across all three lines (Figure S11). No differences were found at later time points (60, 90, or 120 DIV) in COs (Figures S12A-S12C). In contrast, ARX$^+$ cells decreased in

ARX^{PAE} GEOs at 30 and 60 DIV (30 DIV: $p = 0.0033$, d = −1.367; 60 DIV: $p < 0.0001$, d = −1.823; Figures 2H, 2I, and S13 [plots per condition] and S14 [plots per line]), whereas no significant difference was observed at 120 DIV (Figure S12D). Although it has been reported that PAE produces nuclear inclusions and/or cytoplasmatic localization,[28,56-58] ARX was restricted to the nucleus without forming inclusions in COs and GEOs (Figures 2F and 2H, insets). Our data demonstrated that, while ARX is expressed in both cortical and subpallial tissues, PAE affects *ARX* expression levels differently in COs and GEOs.

PAE affects cortical development

The expression of ARX in progenitor cells in COs at 30 DIV and the increased expression of *ARX* in ARX^{PAE} COs prompted us to investigate how PAE affects cortical development. We first analyzed the size of control and ARX^{PAE} COs over time. We found that ARX^{PAE} COs were larger than control COs at 14 and 21 DIV, but no difference was found at 60 DIV (Figure S15; 14DIV: $p = 0.0007$, 21DIV: $p = 0.0226$, 60 DIV: $p = 0.3554$). Interestingly, one of the patients included in the study had a head circumference of 61 cm at 18 years old, corresponding to the 99.99th percentile (*Z* score of 4.07). To obtain a comprehensive profile of cell composition in control and ARX^{PAE} COs, we performed cell-type proportion analysis using the scRNA-seq dataset and observed an increase in the percentage of progenitor cells (RGCs, cycling progenitors, and intermediate progenitors) in ARX^{PAE} COs (Figure 3A). To explore whether the increased size and the higher proportion of progenitors were due to increased proliferation, we labeled Ki67^{+} dividing cells by IHC in sections from control and ARX^{PAE} COs at 30 DIV.[59] We found a significant increase of KI67^{+} cells in ARX^{PAE} COs ($p < 0.008$, d = 0.782; Figures 3B and 3C). A similar result was observed by scRNA-seq (Figures S10 and S11). As KI67 is mostly expressed in cycling progenitor cells (Figures S10 and S11), we then calculated the percentage of cells in each cell cycle phase within this population. Consistent with the increase in Ki67^{+} cells, ARX^{PAE} COs showed a higher percentage of cells in S and G2/M phases with fewer cells in the G0/G1 phase (Figure 3D). Next, we investigated the expression of a selection of 43 genes involved in cell cycle regulation and found upregulation of most of these genes in cycling progenitors in ARX^{PAE} COs (Figure S16). These genes were selected based on their established roles in cell cycle regulation.[60-62] Furthermore, as it has been shown that *Arx* regulates cortical progenitor expansion in mice by repressing *Cdkn1c*, a negative regulator of cell proliferation,[16] we analyzed *CDKN1C* expression in control and ARX^{PAE} COs at 30 DIV by scRNA-seq and RT-qPCR (Figures 3E and S17). While *ARX* expression was increased in ARX^{PAE} COs, consistent with IHC and scRNA-seq results, *CDKN1C* levels were significantly decreased (Figure 3E; *ARX*: $p = 0.0014$; *CDKN1C*; $p = 0.0236$). *CDKN1C* expression was not restricted to a specific cell type (Figure S17A), and most cells within the cortical lineage contributed to the overall reduction of *CDKN1C* observed in ARX^{PAE} COs (Figures S17B and S17C). In addition, we examined the expression of other cell cycle inhibitors, such as *CDKN1A, CDKN1B*, and *RB1.* While their expression levels were lower than that of *CDKN1C*, no significant differences were observed between ARX^{PAE} and control COs (Figures 3E and S17C). These findings align partially with previous mouse studies on ARX, showing that ARX regulates cortical progenitor proliferation, although differences in ARX expression highlight species-specific regulatory mechanisms.

We next examined the developmental dynamics of cortical lineage cells (RGCs, cycling progenitors, intermediate progenitors, and deep-layer neurons) using pseudotime trajectory analysis. We first identified the segment of the trajectory containing these four cell types in both control and ARX^{PAE} COs (Figure S18A). We also observed other partitions that did not include all four types (Figure S18A). Interestingly, in these partitions, most cells, if not all, were ARX^{PAE}, suggesting that some cells fail to differentiate into cortical neurons in ARX^{PAE} COs at 30 DIV. In the partition containing the four cells types (Figure S18B), ARX^{PAE} cells showed a significantly increased distribution toward the endpoint of the trajectory compared to controls ($p < 0.0001$, Kolmogorov-Smirnov test; Figure S18C). Within each cell type, pseudotime analysis revealed a forward shift in ARX^{PAE} COs, with RGCs and IPs showing the most pronounced changes. To further explore these shifts, we investigated the expression of key progenitor markers: PAX6, HES1, and HES5 (hes family basic-helix-loop-helix [bHLH] transcription factors 1 and 5).[40,63,64] ARX^{PAE} COs had significantly fewer PAX6$^+$ cells than controls ($p = 0.0085$, d = 0.955; Figures 3F-3H) and showed a trend toward decreased *HES1* expression (Figures S19A-S19D), while *HES5* expression trended upward (Figures S19A-S19D). No significant differences were observed in SOX2 and VIMENTIN expression at 30 DIV (Figure S19E). Moreover, RT-PCR analysis of *TBR2*, a marker of IPs,[65,66] showed a significant increase in ARX^{PAE} COs (Figure 3I; $p = 0.0123$). In addition, differential gene expression (DGE) analysis of progenitor cells at 30 DIV identified 2,068 differentially expressed genes in ARX^{PAE} COs ($p < 0.05$), with 1,347 upregulated and 721 downregulated (Figure 3J). Among these, 75 upregulated genes had a log fold change (logFC) > 1, and 8 downregulated genes had a logFC < −1. Notably, *TUBB3*, an early neuronal differentiation marker, was among the upregulated genes. RT-qPCR analysis showed a trend toward increased *TUBB3* expression ($p = 0.2224$; Figure 3K) and a significant increase in *MAP2* expression ($p < 0.0001$). Altogether, these results suggest that ARX^{PAE} accelerates cortical development by downregulating early neural genes and upregulating genes associated with later neuronal differentiation.

To investigate the long-term consequences of PAE in the cortex, we analyzed COs at 60, 90, or 120 DIV. No significant marker differences were detected (Figures S20, S21, S22, and S23), though trends suggested fewer PAX6$^+$ cells at 90 DIV and fewer NEUROD1$^+$ neuronal progenitor cells, CTIP2$^+$ and SATB2$^+$ neurons, and FAM107A$^+$ oRGCs at 120 DIV (Figures S21, S22, and 23). Given the absence of major changes in cell type composition at later stages, we next explored whether ARX^{PAE} promotes cell death in cortical progenitors. Cleaved caspase-3 (AC3$^+$) staining did not a reveal significant increase in apoptotic cells in ARX^{PAE} COs compared to controls at any time point, although a trend toward elevated AC3$^+$ cell numbers was observed (Figure S24). Taken together, our data suggest that PAE in the second polyalanine tract initially increases cortical progenitor populations and accelerates their progression toward more differentiated neuronal states.

PAE alters interneuron differentiation and migration

As *ARX* was also expressed in GEOs, we next studied the effect of PAE in GEOs. We first analyzed the size of control and ARX^{PAE} GEOs over time and found that ARX^{PAE} GEOs were larger than controls at all time points and that these size differences were more prominent than in COs (Figure S25; 14 DIV: $p = 0.0291$, 21 DIV: $p = 0.0007$, 60 DIV: $p =$

0.0076). No major differences were found in the number of progenitor cells between control and ARX^{PAE} GEOs at 30 DIV by scRNA-seq (Figure 4A), although there was a slight but significant decrease in $KI67^{+}$ proliferative cells in ARX^{PAE} GEOs at 30 DIV ($p = 0.0494$, $d = -0.672$; Figures 4B and 4C), with no change observed in NKX2.1 expression (Figures 4B and 4C). Although the interneuron cluster was decreased in ARX^{PAE} GEOs at 30 DIV (Figure 4A), no differences were detected in DCX^{+} or $TUJ1^{+}$ cells by IHC (Figure 4D). To further explore the interneuron population, we examined neuronal markers (DCX, TUJ1, and GABA) at later time points. We found a trend toward a decrease in DCX^{+} cells at 60 DIV in ARX^{PAE} GEOs ($p = 0.0575$; Figure 4E) and a significant decrease of $TUJ1^{+}$ at 60 DIV, and $GABA^{+}$ at 90 DIV (TUJ1: $p = 0.0407$; GABA: $p = 0.0433$; Figures 4E and 4F). The most pronounced differences were found at 120 DIV, when ARX^{PAE} GEOs had fewer $TUJ1^{+}$ and $GABA^{+}$ neurons (TUJ1: $p = 0.0368$, $d = -1.196$; GABA: $p = 0.0641$, $d = -0.85$; Figures 4G and 4H). Similar reductions in interneuron populations were also observed in the scRNA-seq data (Figures 4I and S26, plots per condition; Figure S27, plots per line). Although ARX^{PAE} GEOs had fewer interneurons overall, they exhibited higher expression levels of *CALB1*, *CALB2,* and *SST* (Figure S26, plots per condition; Figure S27, plots per line). The number of *PV*-expressing cells was also higher in ARX^{PAE} GEOs compared to controls, although the absolute number of PV^{+} cells was low in both conditions. To determine whether the reduction of interneurons was due to increased cell death, we analyzed the number of $AC3^{+}$ cells by IHC. No significant differences were observed between ARX^{PAE} GEOs and controls at any time point, except for a decrease in $AC3^{+}$ cells at 60 DIV (Figure S28). These data suggest that PAE affects interneuron differentiation during development without altering progenitor cell production.

During development, interneurons are formed in the GE and migrate along circuitous routes to the cortex to integrate with projection neurons and form the cortical neuronal circuits.[51,67] Organoid models can recapitulate interneuron migration by fusing one GEO and one CO (referred to as assembloids). We labeled interneuron progenitor cells in GEOs at 50 DIV using a DLX1/2b GFP-expressing lentivirus (Figure 5A). DLX1 and DLX2 are expressed in ventral progenitor cells and promote interneuron differentiation and migration.[51,68] We also labeled COs using an adeno-associated virus (AAV)-synapsin-mCherry virus; these $mCherry^{+}$ cells did not migrate into GEOs (Figure S29A). After 10 DIV, one GEO was placed in close contact with one CO to allow their fusion and the migration of the DLX1/2-GFP^{+} cells from the GEO to the CO. Migration into the CO was assayed 30 days after fusion by time-lapse imaging. We observed a similar percentage of migrating cells in ARX^{PAE} assembloids (condition 2) compared to controls (condition 1; Figure S29B). However, DLX1/2-GFP^{+} cells migrated farther and faster in ARX^{PAE} assembloids (distance: $p = 0.0424$, $d = 0.607$; velocity: $p = 0.0425$, $d = 0.624$; Figures 5B-5F; Videos S1, S2, S3, and S4). Moreover, in control assembloids, most cells migrated less than 300 μm, whereas a higher percentage of cells migrated more than 300 μm in ARX^{PAE} assembloids (Figure 5G). To determine whether the accelerated migration observed in ARX^{PAE} assembloids was due to premature migration, we analyzed the percentage of mobile vs. immobile DLX1/2-GFP^{+} cells at an earlier time point (75 DIV). Although a smaller percentage of mobile cells was observed at 75 DIV compared to 90 DIV, no differences were found between ARX^{PAE} and control assembloids at this earlier stage (Figure S29C). To investigate whether faster migration of interneurons leads to an increased presence of neurons in the cortex, we quantified the GFP^+ area in the CO and GEO sides of the assembloids. Both ARX^{PAE} and control assembloids showed similar percentages of GFP^+ area (Figure S29D). After migration into the COs, DLX1/2-GFP^+ cells from both control and ARX^{PAE} assembloids expressed GABA (Figure S30A), but they did not express CB or CR (Figure S30B). To determine whether the accelerated migration observed in ARX^{PAE} assembloids was due to a cell-autonomous effect, we fused ARX^{PAE} GEOs to control COs (condition 3) to create inter-individual assembloids in which interneurons carried PAE mutations but cortical cells expressed wild-type ARX (Figures 5D-5G). DLX1/2-GFP^+ cells in these assembloids migrated similar distances and at similar velocities as those in ARX^{PAE} assembloids (Figures 5D-5G). To test whether PAE in cortical cells alone could affect interneuron migration, we fused control GEOs to ARX^{PAE} COs (condition 4). Interneurons in this condition migrated at distances and velocities similar to those observed in control assembloids (Figures 5D-5G). These data suggest that PAE enhances interneuron migration in a cell-autonomous manner.

As CXCR4 is important for neuronal migration[69] and a known target of ARX, we examined whether DLX1/2-GFP^+ cells expressed CXCR4. We found that DLX1/2-GFP^+ cells expressed CXCR4 both before and after migrating into the COs in control and ARX^{PAE} assembloids at 90 DIV (Figure 6A). Next, we determined whether CXCR4 expression was altered by PAE. CXCR4 expression was slightly elevated at 60 and 90 DIV in ARX^{PAE} GEOs (Figures S30C and S30D), but the most significant difference was observed at 120 DIV, as shown by IHC ($p = 0.0037$, d = 1.313; Figures 6B and 6C) and supported by scRNA-seq analysis (Figure S26, plots per condition; Figure S27, plots per line). While CXCR4 expression declined over time in control GEOs, it remained stable in ARX^{PAE} GEOs (Figure S30). Given that ARX also regulates *CXCR7*, we evaluated whether *CXCR7* expression was similarly affected by the PAE mutation. Although *CXCR7* levels were generally lower than *CXCR4*, no significant differences were observed between control and ARX^{PAE} GEOs (Figures S26 and S27). These results prompted us to explore whether CXCR4 inhibition is sufficient to reduce the enhanced migration observed in ARX^{PAE} assembloids. Following a 15-h baseline recording, control and ARX^{PAE} assembloids were treated with AMD3100, a CXCR4 inhibitor, and interneuron migration was evaluated for an additional 15 h (Figure 6D). AMD3100 did not affect interneuron migration in control assembloids but significantly reduced migration in ARX^{PAE} assembloids to control levels ($p = 0.0024$; Figure 6E). Overall, our results indicate that PAE alters interneuron differentiation and promotes migration, a phenotype that can be reduced by CXCR4 inhibition.

Impact of PAE on neuronal activity

We next investigated whether the cellular and molecular alterations observed in COs and GEOs derived from patients with PAE mutations in the *ARX* gene impact neural network connectivity and functionality. We performed multi-electrode array (MEA) recordings in assembloids at 120 DIV, a developmental time point shortly after the observed interneuron migration abnormalities (Figure 7A). ARX^{PAE} assembloids exhibited a significantly higher number of active electrodes and increased spike counts compared to controls (mean firing rate [MFR]: $p = 0.00020$, active electrode: $p = 0.0055$; two-way ANOVA, genotype effect;

Figures 7B-7E and S31). Synchronous burst activity was detected in a subset of samples in both groups (3 of 6 in controls, 5 of 6 in ARX^{PAE}), though this was not statistically analyzed due to the small sample size. To investigate the contribution of GABAergic signaling to network activity, we applied GABA and picrotoxin, a $GABA_A$ receptor antagonist, to the assembloids. Surprisingly, neither compound significantly altered spiking or bursting activity in either genotype at 120 DIV (Figures 7B-7E and S31). These data suggest that GABAergic signaling remains excitatory at this stage, consistent with early developmental physiology. Collectively, our data suggest that ARX^{PAE} assembloids exhibit increased network excitability, potentially driven by altered interneuron development and immature inhibitory signaling.

Our findings highlight the value of neural organoids as a model for studying epilepsy and neurodevelopmental disorders. We demonstrate that PAE mutations in ARX exert context-dependent effects on different neural cell types. In cortical cells, PAE mutations enhance progenitor proliferation, whereas in interneurons, they impair differentiation and migration from the subpallium. These disruptions in neural development lead to altered neuronal activity, which may contribute to the seizure phenotypes observed in patients with PAE mutations.

DISCUSSION

ARX is a transcription factor critical for brain development.[14,16,17] To study the impact of PAE mutations in *ARX*, we used patient-derived hiPSCs to generate COs and GEOs. We found that ARX disruption affects both brain areas through distinct mechanisms: PAE promotes cell proliferation in the cortex (COs and dorsal progenitor cells) while enhancing interneuron migration. These findings were consistent across lines, suggesting robust, mutation-driven mechanisms despite the lack of isogenic controls. Our findings provide insight into PAE pathology and establish a human model for therapeutic development.

ARX is expressed in both the developing and adult brain.[14,15] During development, *Arx* is expressed in dorsal progenitor cells, while in adults its expression is restricted to interneurons originating from the GE.[14,16,17] In this study, we found ARX expression in both proliferating progenitor cells and differentiated cortical neurons in COs. Consistent with murine studies, ARX was also expressed in ventral progenitor cells and interneurons in GEOs. Moreover, our analysis of publicly available brain transcriptome data revealed *ARX* expression in both human embryonic and postnatal cortical and subpallial tissues, highlighting its widespread role during brain development across dorsal and ventral compartments. Our data also demonstrate that PAE mutations affect *ARX* expression in a context-dependent manner, varying by cell type and developmental timing. For instance, we observed an increase in the number of ARX$^+$ cells in COs, suggesting a gain-of-function effect, while in GEOs, ARX expression was decreased at early time points, suggesting a loss-of-function effect. This observation contrasts with findings from mouse models, where reductions in ARX protein levels have been reported in embryos harboring PAE mutations in either the first or second PA tract.[19,70] Similarly, adult mice with PAE mutations in the first tract showed reduced numbers of ARX$^+$ cells in the cortex, hippocampus, and striatum.[28] These differences suggest that PAE mutations may impact *ARX* expression differently in humans compared to mice. Additionally, while nuclear aggregates of overexpressed ARX with PAE mutations have been reported *in vitro*,[56,57] they have not been observed *in vivo*,[19,28,70] suggesting that, under physiological conditions, ARX does not form nuclear inclusions. Cytoplasmic Arx expression has also been observed in mouse models.[28,58] Interestingly, we did not detect nuclear inclusions or cytoplasmic ARX expression in either COs or GEOs, suggesting that a different mechanism may underlie PAE pathology in humans. Further analysis of ARX expression across human brain regions and developmental stages will be necessary to better understand PAE's impact and to guide novel therapeutic strategies for ARX-related neurodevelopmental disorders.

RGCs are the neural stem cells of the cortex, arising from the neuroepithelium and differentiating into neurons.[71,72] In our study, we found that PAE in *ARX* led to an increase in cortical progenitor cells along with increased *ARX* expression in COs at 30 DIV. Pseudotime analysis revealed an accelerated developmental trajectory of cortical progenitor cells in ARX^{PAE} COs, supported by a mixed gene expression profile. Although canonical neural progenitor markers such as *MKI67* (a proliferation marker) and *HES5* were upregulated, other key progenitor genes, such as *PAX6* and *HES1*, were downregulated. Moreover, these COs exhibited elevated expression of neuronal genes such as *TUBB3* and *MAP2ab*, suggesting premature neuronal differentiation likely caused by PAE. Our findings partially align with previous mouse studies, which demonstrated that *Arx* KO mice exhibited thinner cortices and fewer bromo-deoxyuridine (BrdU)$^+$ cells.[15] Furthermore, deletion of *Arx* from pallial progenitor cells reduced the number of BrdU$^+$, Pax6$^+$, and Tbr2$^+$ cells while increasing Tuj1$^+$ cells.[16] Similarly, downregulation of *Arx* in cortical progenitors led to premature cell cycle exit, while *Arx* overexpression extended the progenitor cell cycle length.[17] Notably, *Arx* KO mice showed upregulation of *Cdkn1c*,[16] and in our study, we observed a reduction of *CDKN1C* levels in ARX^{PAE} COs at 30 DIV, coinciding with increased *ARX* expression. This suggests that *CDKN1C* may be a direct downstream target of ARX and could mediate the proliferation phenotype observed in ARX^{PAE} COs. Interestingly, similar accelerated developmental trajectories have been described in organoid models containing mutations in ASD risk genes,[73] suggesting potential shared mechanisms across neurodevelopmental disorders.

The increased cell proliferation observed could explain the larger size of ARX^{PAE} COs at 14 and 21 DIV as well as the macrocephaly reported in some patients.[74] However, it is important to note that not all progenitor cells will successfully differentiate into neurons; some may undergo apoptosis, as suggested by the trend toward increased AC3 staining in ARX^{PAE} COs at 30 DIV and the absence of significant phenotypes at later time points. Since cell death is cumulative, and apoptotic cells are removed during medium changes, even small increases in AC3$^+$ cells may reflect substantial cell loss over time. While neuronal cortical defects specifically related to PAE mutations have not been extensively studied in mouse models or patients, disorganization of pyramidal neurons has been documented in conditions such as X-linked lissencephaly with abnormal genitalia (XLAG)[75] and in *Arx* KO mice.[15] Moreover, reductions in upper-layer cortical neurons have been reported in postnatal *Arx* cKO mice.[16] Although we did not detect a significant difference in the total number of cortical neurons in our study, we observed a slight decrease in CTIP2$^+$ and SATB2$^+$ neurons in ARX^{PAE} COs at 120 DIV. Future studies will be needed to fully elucidate the role of ARX in cortical neuron development.

A reduction in GABAergic interneurons has been observed in the cortex of both *Arx* KO mice and PAE mouse models[19-22,25,28] as well as in zebrafish larvae.[76] Similarly, patients with XLAG exhibit reduced numbers of interneurons.[75] Given that interneurons originate in the GE and migrate tangentially to the cortex, these findings suggest that ARX plays a crucial role in regulating neuronal migration.[51,67] Supporting this idea, previous studies have shown the accumulation of cells within the GE following *Arx* inactivation or overexpression,[15,17,25] and aberrant migration has been demonstrated in *Arx* KO forebrain slice cultures.[18] Notably, Lee et al. reported that, while PAE in the first tract inhibits neuronal migration, PAE mutations in the second tract does not affect migration.[25] However, our results show that human PAE mutations in the second tract initially compromise GABAergic differentiation and subsequently enhance interneuron migration to the cortex. Using time-lapse imaging, we observed that DLX1/2-GFP$^+$ interneurons carrying PAE mutations exhibited accelerated migration, moving farther and faster into cortical tissue in a cell-autonomous manner, as shown by inter-individual assembloid experiments. Despite this increased migration, we did not observe an increased number of interneurons in the cortex. This discrepancy suggests several possibilities: interneurons with PAE mutations may continue migrating due to failure in receiving stop signals or locating their final targets, or they may fail to properly integrate into cortical circuits, ultimately leading to cell death. Further experiments, such as retrograde tracing using rabies virus, will be necessary to determine how PAE expansion in ARX affects interneuron connectivity after migration. Notably, we observed increased expression of CXCR4 in GEOs and of its ligand CXCL12 in COs. The CXCR4/CXCL12 signaling axis plays a critical role in interneuron migration.[53,54,69,77] Previous studies have shown that blocking CXCR4 decreases interneuron migration in human assembloids.[32] Importantly, CXCR4 is a transcriptional target of ARX, with ARX promoting its expression.[22,78] It is therefore conceivable that PAE mutations enhance ARX binding to the CXCR4 promoter, leading to increased CXCR4 expression and accelerated migration. Consistent with this idea, treatment with AMD3100, a CXCR4 inhibitor, reduced interneuron migration in ARX^{PAE} assembloids to control levels but had no significant effect on migration in control assembloids. The lack of response to AMD3100 in controls could reflect the naturally lower expression of CXCR4 at later developmental stages. Interestingly, while CXCR4 expression declined over time in control assembloids, it was maintained in ARX^{PAE} assembloids, suggesting a prolonged migration phase in patient-derived cells. These findings suggest that, although additional mechanisms may contribute,[79,80] PAE mutations in ARX primarily promote interneuron migration through upregulation of CXCR4 signaling.

Patients with PAE mutations in *ARX* frequently develop epilepsy, among other neurological manifestations.[81,82] In mouse models of PAE,[20,21] it has been proposed that defects in interneuron number or function are the primary cause of seizure development, while cortical abnormalities contribute to other behavioral phenotypes.[24] Human organoid models, which display complex neuronal oscillations resembling early human brain activity, offer a unique opportunity to model cortical function in disease states.[38,83] To investigate the impact of PAE mutations on neuronal network function, we compared MEA recordings from control and ARX^{PAE} assembloids at 120 DIV. Our data revealed heightened network activity in ARX^{PAE} assembloids, as evidenced by an increased spike number compared to controls at 120 DIV. ARX^{PAE} assembloids also exhibited broader network activation with a greater number of active electrodes recorded across all conditions. Both controls and ARX^{PAE} assembloids showed only subtle responsiveness to 20 μM GABA at 120 DIV, suggesting that $GABA_A$ receptor-mediated signaling had not yet fully switched from excitatory to inhibitory at this developmental stage. While these results indicate early hyperactivity in ARX^{PAE} assembloids, they reflect an immature stage of disease progression. Longer-term MEA recordings and analysis of older assembloids will be necessary to fully evaluate the potential for epilepsy-like network behavior in more mature systems. Additionally, the excitatory and inhibitory receptors, such as α-amino-3-hydroxy-5-methyl-4-isoxa-zolepropionic acid (AMPA), N-methyl-D-aspartate (NMDA), $GABA_A$, and $GABA_B$, should be specifically targeted using selective agonists and antagonists to gain a deeper understanding of how ARX mutations affect neural network development. Furthermore, since ARX is also expressed in other brain regions, including the hippocampus, striatum, and thalamus,[14,15,84] current organoid models lacking these structures may not fully capture the epileptic phenotype observed in patients with PAE mutations. In fact, Beguin et al. proposed that epileptic activity originates in the hippocampus in *Arx* mouse models.[85] Developing more complex organoid systems that incorporate multiple brain regions will be critical for studying diseases involving widespread neurodevelopmental defects and for disentangling the regional contributions to the overall phenotype.

Our results demonstrate that ARX plays a critical role during brain development. PAE mutations in its second tract expand cortical progenitors, accelerate differentiation, and enhance interneuron migration via altered CXCR4 signaling. These defects in cortical and GABAergic neurons lead to early network hyperactivity, providing mechanistic insight and a human model for therapy development.

Limitations of the study

Using COs and GEOs derived from controls and patients carrying the PAE mutation in *ARX*, we demonstrated distinct mechanisms affecting each brain region. While these findings provide new insights into PAE-driven pathology, several limitations remain.

1. Isogenic controls. The absence of genetically matched controls limits exclusion of confounding factors.
2. ARX expression analysis. Compatible antibodies or ARX-tagged hiPSC lines are needed to fully characterize ARX expression dynamics.
3. Premature cortical differentiation. Pseudotime and bio-informatic analyses suggest accelerated progenitor differentiation, but lineage-tracing experiments (e.g., BrdU) are required for confirmation.
4. Cortical neuron development. Despite increased progenitor proliferation, no significant changes in total cortical neuron numbers were observed; further studies are needed.

5. Interneuron integration. Accelerated migration in ARX^{PAE} assembloids warrants additional connectivity assays (e.g., rabies virus-based retrograde tracing).
6. Organoid maturation. Current models reflect early development; more advanced systems are necessary to better mimic patient phenotypes.

STAR★METHODS

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

We recruited three healthy controls and three patients with eight additional alanines in the second poly-alanine tract of ARX. Information about gender, age, genotype and a brief clinical summary for each line can be found in Table S1. Only male patients were included in the study because ARX is located on the X chromosome, and females typically present with milder and more variable phenotypes. Approval for the study was obtained from the University of Texas at San Antonio IRB panel (#18-146R), and informed consent was obtained from all participants. All lines were authenticated, confirming expression of pluripotency markers, absence of reprogramming factors, normal karyotype, absence of mycoplasma, and length of the second poly-alanine tract of *ARX* (Figure S9). hiPSCs were cultured in mTeSR1 medium (Cat. No. 05851, Stemcell Technologies) on six-well plates (Cat. No. 3506, Corning) coated with growth factor-reduced Matrigel (Cat. No. 356230, BD Biosciences) at 37° C and 5% CO_2. During passaging, ROCK inhibitor (10 μM final concentration; Cat. No. S-1049, Selleck Chemicals) was added to the medium. The medium was replaced daily without ROCK inhibitor until cells reached approximately 70% confluency. Cells were then passaged at a 1:10 ratio using Versene solution (Cat. No. 15040-066, Thermo Fisher Scientific). Power analysis was used to determine the sample size prior to analysis.

METHOD DETAILS

Pluripotent stem cell generation and maintenance: A total of six human induced pluripotent stem cell (hiPSC) lines were generated from peripheral blood samples collected from three healthy control subjects and three ARX^{PAE} patients (Table S1) using either the Invitrogen episomal reprogramming kit or the CytoTune-iPS Sendai Reprogramming kit (Thermofisher). Two clones per line were used in all downstream experiments. Pluripotency was confirmed by immunocytochemistry and flow cytometry, and all lines maintained a normal karyotype and were free of mycoplasma contamination throughout the study (Figure S9). Absence of reprogramming vector integration was also confirmed. hiPSCs were maintained in mTeSR1 medium (Cat. No. 05851, Stemcell Technologies) on six-well plates (Cat. No. 3506, Corning) coated with growth factor-reduced Matrigel (Cat. No. 356230, BD Biosciences). For passaging, ROCK inhibitor (final concentration 10 μM, Cat. No. S-1049, Selleck Chemicals) was added to the medium. Media was changed daily without ROCK inhibitor until cells reached 70% confluency. Cells were passaged at a 1:10 ratio using Versene solution (Cat. No. 15040-066, Thermo Fisher Scientific).

Genotyping: Poly-alanine expansion in the second tract of *ARX* was confirmed by Sanger sequencing. Genomic DNA was extracted from hiPSCs using the DNeasy Blood & Tissue Kit (QIAprep, Cat. No. 69504) following manufacturer's instructions. OneTaq

Hot Start DNA Polymerase (New England BioLab, Cat. No. M0481) was used to amplify the second alanine tract of *ARX* using forward (GGGGGCCGCCTCCTTCAG) and reverse (CCTGGTGAAGACGTCCGGGTAGTG) primers. PCR conditions were as follow, 94° C for 4 min; 94° C 30 s, 60° C 30 s, 68° C 1 min for 35 cycles; 68° C 5 min. PCR products (798 bp) were visualized on an agarose gel and submitted for Sanger sequencing.

Flow cytometry analysis: To analyze pluripotent stem cell markers, hiPSCs were dissociated using accutase (Sigma). 10^6 cells were resuspended in D-PBS/1% fetal bovine serum (FBS). After two washes with D-PBS/1% FBS, cells were incubated for 20 min on ice with primary antibodies (anti-SSEA-4 PE antibody, 1:20, Stemgent Cat. No. 09-0003 or anti-TRA1-60 PE, 1:5, BD Cat. No. 560193). Then, they were washed three times and resuspended in 300 μL D-PBS/1% FBS and analyzed by flow cytometer (BD FACSAria) with 10,000 events per determination. Post-acquisition analysis was done using Flowjo software (Tree Star Inc) and the percentage of cells expressing each marker was determined.

Organoid culture: Human cortical organoids (COs) and human ganglionic eminence organoids (GEOs) were generated based on the protocol established by Pasca and colleagues, with slight modifications.[32,35] Briefly, hiPSCs were dissociated using Accutase (Sigma), and 9,000 cells were resuspended in 150 μL of neural induction medium (DMEM/F12, 20% knock-out serum replacement, Glutamax, MEM-NEAA, 0.1 mM 2-mercaptoethanol, and penicillin/streptomycin) supplemented with the ROCK inhibitor Y27632 (20 μM). Medium was changed daily. From days 1–5, dual SMAD inhibitors was applied using dorsomorphin (5 μM, Sigma) and SB-431542 (10 μM, Tocris). From days 6–24, cells were cultured in neural medium (Neurobasal A, B-27 without Vitamin A, Glutamax, and penicillin/streptomycin) supplemented with bFGF (20 ng/mL, Peprotech) and EGF (20 ng/mL, Peprotech). On day 25, organoids were transferred to neural medium containing BDNF (20 ng/mL, Peprotech) and NT-3 (20 ng/mL, Peprotech) and maintained in this medium until day 42. For GEO specification, additional patterning factors were added: the Wnt inhibitor IWP-2 (5 μM, Selleckchem) from days 4–22 and the Smo pathway activator SAG (100 nM, Selleckchem) from days 12–22 (Figure 1A). From day 25 onward, organoids were transferred to an orbital shaker to improve oxygenation. Organoid were cultured individually in 96-well plates until day 24, then transferred to 24-well plates for long-term culture.

Interneuron migration: On day 50, GEOs were infected with a lentivirus hDlx1/2b:GFP (a gift from Dr. John Rubenstein). and CO were infected with AAV-DJ1-hSYN1:mCherry (Stanford Gene Vector and Virus Core, Stanford University School of Medicine). On day 60, one GEO and one CO were paired in a well of a 24-well plate, placed in close contact in neural medium, and incubated without disruption for 3-4 days to allow fusion. At day 90, fused assembloids were transferred to glass-bottom plates (Corning), and the migration of hDLX1/2b:GFP interneurons into the cortical region was recorded under environmentally controlled conditions (37° C, 5% CO_2). Time-lapse imaging was performed using a Leica TCS SPE8 confocal microscope with a motorized stage. z stack images were captured every 3 μm over a depth of 150–200 μm thickness, with one image taken every 20 min for a total duration of 15 h (Figure 5A). Only neurons that remained in the field of view for the full recording period were included in the analysis. Assembloids with significant drift were excluded. In some experiments, following the initial 15-h time-lapse, assembloids were treated with AMD3100 (a CXCR4 inhibitor; 100 nM; TOCRIS Cat. No. 3299) and imaging was continued for an additional 15 h to assess drug effects on migration (Figure 6D). Selected assembloids were fixed, sectioned, and imaged post-hoc. Neuronal migration distance and average velocity of hDLX1/2b:GFP^{+} cells were quantified using ImageJ. Videos processing was performed with Leica imaging software.

Immunohistochemistry: Organoids were fixed overnight at 4° C in 4% paraformaldehyde and then incubated in 30% sucrose for 48 h at 4° C. After cryoprotection, organoids were embedded in OCT compound and frozen. Sections (14-µm thick) were cut using a cryostat and mounted on slides. For immunohistochemistry, slides were incubated in blocking solution (0.3% Triton X-100, 3% normal donkey serum in TBS) for 1 h at room temperature, followed by incubation with primary antibodies (Table S2) overnight at 4° C in a humidified chamber. After three washes in TBS, slides were incubated with secondary antibodies (Jackson Immunoresearch, 1:500) for 2 h at room temperature. Nuclei was counterstained with 4′,6-diamidino-2-phenylindole (DAPI; Sigma, Cat. No. D9542), and slides were coverslipped using polyvinyl alcohol solution (PVA; Sigma, Cat. No. BP168-122). Fluorescent images were acquired using a Leica SPE8 confocal microscope with identical settings for control and ARX^{PAE} organoids. Four sections per organoid were imaged. Fluorescence intensity area was quantified using ImageJ software.[86-89] A consistent fluorescent threshold was applied across control and ARX^{PAE} samples for each marker. Because some markers were not uniformly distributed, quantification was performed over the entire section. For nuclear markers, data were expressed as the percentage of marker-positive area relative to DAPI area. For cytoplasmic markers, data were expressed as the percentage of marker-positive area relative to the total section area. Each data point represents the mean value from four sections per organoid. For most analyses, three organoids per clone per line per condition were used (total of 18 organoids per condition). All analyses were performed blinded to genotype. To validate this method, we conducted an analysis using nuclear markers and compared manual cell counting with our approach. Regions of interest (ROIs) were selected from eight images stained for PAX6 or CTIP2 (four each), including both control and patient-derived COs. We then performed a correlation analysis between the results obtained using both methods, which revealed a strong correlation ($r = 0.9624$, $p = 0.0001$), supporting the reliability of our methodology (Figure S32).

RNAscope: RNAscope was performed on cryosections following manufacturer’s instructions (ACDBio), including an antigen retrieval step. Custom-designed probes targeting human *HES1* and *HES5* mRNA were obtained from ACDBio. Four sections per organoid were imaged using a Leica SPE8 confocal microscope with a 40× oil immersion objective. Quantification was performed using CellProfiler software, and data are shown as the ratio of *HES1* or *HES5* mRNA signal to DAPI fluorescence.

RNA isolation, RT-PCR and quantitative PCR: RNA was isolated from organoids using the RNeasy Micro Kit (Qiagen) according to manufacturer instructions. The concentration and purity of the RNA samples were measured using a NanoDrop spectrophotometer (Thermo Fisher Scientific). The extracted RNA (500 ng) was reverse transcribed according to the protocol supplied with SuperScriptIII First-Strand Synthesis System for RT-PCR (Cat. No. 18080-051, Invitrogen Life Technologies). Quantitative real-time PCR (qRT-PCR) was carried out using ViiA (Applied Bio-systems, Foster City, CA, USA) using PowerUp SYBR Green Master Mix for qPCR (Cat. No. A25741, Applied Biosystems). Reactions were run in triplicate and expression of each gene were normalized to the geometric mean of GAPDH as a housekeeping gene and analyzed by using the ΔΔCT method. Primer sequences are listed in Table S3.

Dissociation of organoids and single cell RNA sequencing (scRNA-seq): For single-cell RNA seq, COs and GEOs were dissociated using the Worthington papain dissociation kit (Cat. No. LK003150) per the instruction manual. Each organoid subtype (CO or GEO) and timepoint (30 DIV and 120 DIV) was processed independently. One clone per line from three different lines was used per condition. Briefly, 3–4 organoids were placed in a Petri dish with 5 mL of pre-warmed papain solution and 250 μL of DNase. Organoids were minced using sterile blades and the dishes were incubated on a digital rocker at 37° C for 60 min. Digested tissue was transferred to a 15-mL tube, missed with 5 mL of room temperature Earle’s Balanced Salt Solution (EBSS), and gently triturated using successive pipetting with 10mL, 1mL and 200-μL pipettes. Undissociated tissue was allowed to settle at the bottom of the tube, and the cloudy suspension was transferred to a new tube. A 3.15 mL inhibitor solution (2.7 mL of EBSS, 300 μL of reconstituted Worthington albumin-ovomucoid solution and 150 μL of DNase solution) was layered over the cloudy suspension, followed by centrifugation at 300*g* for 5 min at room temperature. The supernatant was discarded, and the resulting cell pellet was suspended in 500 μL of cold Neurobasal A media. Cell viability was determined using trypan blue exclusion.

Single-cell libraries were prepared by the University of Texas at San Antonio (UTSA) Genomics Core using the 10x Genomics Chromium platform. Cell suspensions were loaded into 10x Chromium microfluidics chips (Chip B, Cat. No. 1000153) to generate Gel Beads-in-emulsion (GEMs), targeting 10,000 cells/sample. Libraries were constructed using 10x Genomics 3′ Gene Expression Reagent kit v3 (Cat. No. 1000269), following manufacturer’s recommendations. Sequencing was performed to a depth of ~50,000 reads per cell using either the Illumina NextSeq 500 (at UT Health San Antonio) or the NovaSeq 6000 (at the North Texas Genome Center). Samples were processed in matched batches: Control 1 and $ARX^{PAE}1$, Control 2 and $ARX^{PAE}2$, and Control 3 and $ARX^{PAE}3$ were prepared and sequenced together to minimize batch effects. Although batch correction was applied, residual batch effects were still observed, limiting gene expression comparisons to within-batch analyses.

scRNA-seq data analysis: Raw sequencing data were processed using Cell Ranger (10x Genomics) and Seurat, as described below. Base call files were converted to FASTQ reads using the cellranger mkfastq function. Reads were aligned to the GRCh38–3.0.0 human reference genome assembly and cell-by-gene count matrices were generated using the cellranger count function with default parameters, except ‘–expect-cells’ set to 8,000.

Libraries for Control 1 and ARX^{PAE} 1 at 120 DIV (COs) were re-sequenced to increase read depth and aggregated using Cell Ranger's function. Subsequent analysis was performed using Seurat v.3.2 in (R v.3.6.0), Seurat v4.0 (R v.4.0.3), and Seurat v5.2 (R v.4.1.1).[90,91] For each developmental timepoint (30 DIV and 120 DIV) and organoid subtype (CO and GEO), SeuratObject were created separately for each sample (Control 1–3) and ARX^{PAE} 1–3) from the filtered feature-barcode matrices. Genes expressed in ≥10 cells (min.cells = 10) and cells with ≥500 detected genes (min.features = 500) were retained. The six SeuratObjects were merged using Seurat's merge function. Further quality control filtering retained cells with nFeature_RNA>500 and nFeature_RNA<6000. Mitochondrial and ribosomal gene percentages were calculated and later regressed out during normalization. Unique molecular identifier (UMI) counts were normalized to transcripts per million (TPM x 10^6) and log-transformed. Variable genes were identified using the mean.var.plot method, and the ScaleData function was used to regress out mitochondrial and ribosomal gene contributions. Dimensionality reduction was performed using principal component analysis (PCA). The top 19 principal components (PCs) were selected on Seurat's ElbowPlots function. Batch effects were corrected using the RunHarmony function (Harmony v.0.1.0). Clustering was conducted in the Harmony-reduced PCA space using Seurat's FindNeighbors function with reduction = 'harmony' and dims = 1:30 followed by FindClusters with resolution = 0.5. Uniform manifold approximation and projection (UMAP) or t-distributed stochastic neighbor embedding (tSNE) in Seurat was finally used as a non-linear dimensionality reduction technique to visualize and further explore the cells in each dataset. The number of cells sequenced and passing QC included in each dataset are presented in Table S4.

Upregulated genes in each cluster were identified using the VeniceAllMarker tool from the Signac package (v.1.3.0) from BioTuring (https://github.com/bioturing/signac). Clusters were annotated based on the top 20 most highly expressed genes, which were compared to published scRNA-seq datasets from human fetal brain and organoid studies).[38,92,93] If the top markers could not unambiguously define a cluster, it was labeled as "Undefined". Heatmaps of cell type-defining genes were generated using the DoHeat-map function in Seurat. Cell type proportions were estimated using the glmer function from the lme4 package (v.1.1.27.1), adapted from Paulsen et al.[73] Pseudobulk differential gene expression analysis in progenitor populations was performed using Seurat's AggregateExpression() function. Volcano plot were generated using the EnhancedVolcano package (v.1.8.0) in R. The scRNA-seq data are available from the NIH Gene Expression Omnibus (GEO) under accession number GSE215362. The code used for data analysis is available on GitHub: https://github.com/parulvarma123/ARX_Organoids_scRNAseq. https://doi.org/10.5281/zenodo.17426624.

Cell cycle analysis: Cell cycle phase scores for cycling progenitors were calculated using the CellCycleScoring() function in Seurat, which assigns a score based on expression of S and G2/M phase gene markers. Cells not expressing markers of either phase were classified as non-cycling (G1 phase).

Pseudotime analysis: Pseudotime analysis was performed using Monocle3 v1.3.7.[94] The pipeline was adapted from Paulsen et al.[73] A new Seurat object was created containing radial glial cells, cycling progenitors, intermediate progenitors and deep-layer neurons from COs at 30DIV. A (cell_data_set) (CDS object) was created from this subset, ensuring equal numbers of control and ARX^{PAE} cells. The CDS was preprocessed with preprocess_cds(), followed by dimensionality reduction (reduce_dimensions) and clustering (cluster_cells), and partitioned datasets. A partition that included all cell types was chosen for trajectory inference using learn graph(). The root note for pseudotime ordering was manually assigned to radial glial cells. Statistical comparisons of pseudotime distributions between control and ARX^{PAE} cells were performed using the Kolmogorov-Smirnov test.

Multi-electrode array (MEA) assay: Electrophysiological recordings were carried out inside a humidified incubator (95% RH, 37° C and 5% CO_2). To improve maturation, the organoids used for MEA recordings were cultured in Neurobasal Plus medium (Thermo Fisher Scientific, Cat. No. A3582901) supplemented with B27 Plus (Thermo Fisher Scientific, Cat. No. A3582801) post fusion. For each recording session, organoids were transferred to 60-electrode 2D (60MEA200/30iR-Ti) or 3D multi-electrode arrays (MEAs) (60-3DMEA200/12/80iR-Ti) from Multichannel Systems and bathed in 200 µL media to ensure adherence without floating. MEA recordings were consistently performed at the junction between the CO and GEO regions of the assembloids, covering electrode pitch. Recordings following sequential conditions: baseline, 20 µM GABA (Sigma-Aldrich, Cat. No. 56-12-2) and 58 µM picrotoxin (Sigma-Aldrich, Cat. No. 124-87-8), with 15-min intervals to data acquisition. Each recording session lasted 30 min. MEAs were sealed with fluorinated Teflon film from Multichannel Systems (MEA-MEM-set). Each 2DMEA contained 60 Titanium nitrate (TiN) microelectrodes (30 µm of diameter, 200 µm inter-electrode pitch, 8 × 8 layout), while 3DMEAs featured 60 TiN microelectrodes (50 µm diameter, 200 µm inter-electrode pitch, 50 µm height). As no differences were found between 2D or 3D recordings, data were combined. Extracellular potentials were amplified (ME2100-Mini, Multichannel Systems), sampled at 25 kHz/channel, digitized at 16-bit resolution, and recorded using Experimenter software for offline analysis. Raw recordings were preprocessed using *SpiQ* in Julia (Giugliano et al., available at Github: https://github.com/mgiugliano/SpiQ).[95,96] Channels were batch-processed in parallel using *crunch.jl*, applying filtering, spike detection, and sorting parameters from a TOML configuration file. Spike detection was performed on band-pass filtered traces (400–3000 Hz), identifying events that exceeded 5× the noise standard deviation, with no upper threshold limit. Both positive and negative crossings were detected, and a 2.5 ms refractory period was enforced. Electrodes were classified as active if they contained any spike events meeting these criteria. Low-frequency components (<100 Hz) were excluded. Spike sorting used the same band-pass range, with waveform extraction windows of 1.0 ms pre-spike and 2.5 ms post-spike. Synchronous spikes were defined as events occurring within a ±0.01 s window across electrodes and were color-coded by condition (green: controls; orange: ARX^{PAE} organoids; gray: others). Firing rates were computed in 0.1 s bins and normalized by electrode count. For spatial mapping, mean firing rates per electrode were calculated over a 30-min recording and projected onto the 8 × 8 MEA layout. Visualizations were generated using custom Python scripts (Python 3.9) with NumPy, Matplotlib, Seaborn, and ipywidgets.

Human fetal brain tissue analysis: Bulk RNA-seq data for the human fetal brain were downloaded from the BrainSpan Atlas of the Developing Human Brain (https://brainspan.org/). RPKM (reads per kilobase transcripts per million mapped reads) values for *ARX* expression as normalized using the following formula RPKM(gene)/(∑RPKM(all genes in the sample)) X 10^6. Data were plotted using Graphpad Prism, spanning 8 weeks post-conception (wpc) to 40 years of age.

QUANTIFICATION AND STATISTICAL ANALYSIS

Outliers were removed using GraphPad Prism (Q = 1%). For normally distributed data, two-tailed Student's t test were used to compare the mean ± standard error of the mean (SEM) values, with Welch's correction applied when variances were unequal (per F-test). Mann-Whitney U tests were used for non-parametric comparisons. Paired t-tests were used for migration experiments involving AMD3100. two-way ANOVA was used for MEA recordings. Details on statistical tests and sample sizes are provided in the figure legends and supplemental tables. Statistically significant was set at $p < 0.05$. Effect size (*Cohen's d*) was calculated using: https://www.psychometrica.de/effect_size.html., with d ≥ 0.8 considered a large effect.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Highlights

- Neural organoid model for ARX-associated epilepsies
- ARX^{PAE} mutations promote cortical progenitor proliferation
- Interneuron migration is enhanced in ARX^{PAE} mutations through CXCR4 signaling
- ARX^{PAE} assembloids show early network hyperactivity

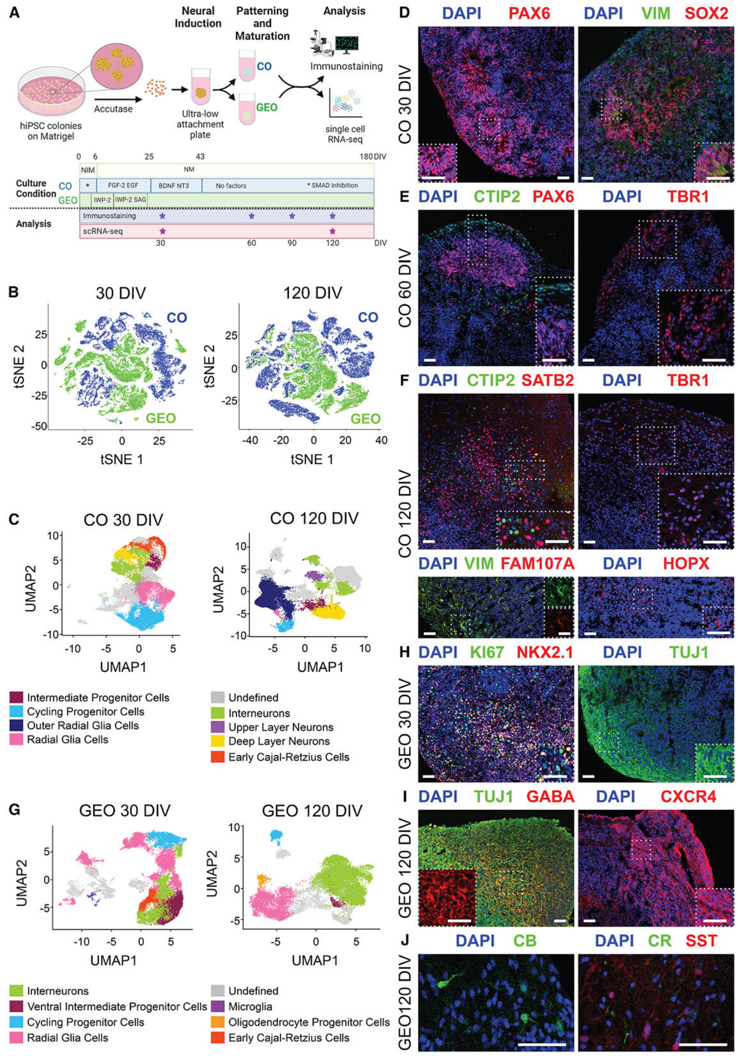

Figure 1. Generation and characterization of human organoids

(A) Experimental design and culture conditions used to generate COs and GEOs from hiPSCs.

(B) t-distributed stochastic neighbor embedding (tSNE) plots comparing control COs and GEOs at 30 and 120 DIV.

(C) Uniform manifold approximation and projection (UMAP) plots of control COs at 30 and 120 DIV.

(D) Control COs at 30 DIV immunostained for PAX6, VIMENTIN, and SOX2 and counterstained with DAPI.

(E) COs at 60 DIV immunostained for PAX6, CTIP2, and TBR1 and counterstained with DAPI.

(F) COs at 120 DIV immunostained for CTIP2, SATB2, TBR1, VIMENTIN, FAM107A, and HOPX and counterstained with DAPI.

(G) UMAP plots of GEOs at 30 and 120 DIV.

(H) GEOs at 30 DIV immunostained for KI67, NKX2.1, and TUJ1 and counterstained with DAPI.

(I) GEOs at 120 DIV immunostained for TUJ1, GABA, and CXCR4 and counterstained with DAPI.

(J) GEOs at 120 DIV immunostained for CB, CR, and SST and counterstained with DAPI. Scale bar: 50 μm. DIV, days *in vitro*; CB, calbindin; CR, calretinin; CO, cortical organoid; GEO, ganglionic eminence organoid; NIM, neural induction medium; NM, neural medium; SST, somatostatin; VIM, VIMENTIN. $n = 18$ organoids per condition (2 clones × 3 lines).

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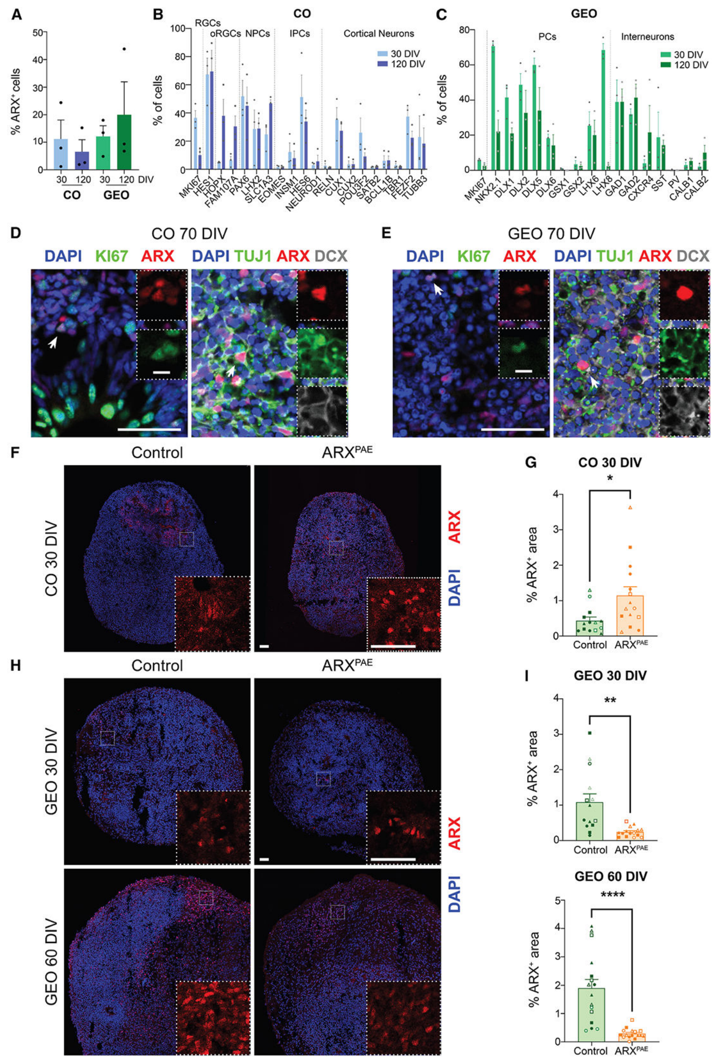

Figure 2. ARX expression in COs and GEOs

(A) Percentage of ARX-expressing cells (ARX$^+$ cells/total cells) in control COs and GEOs at 30 and 120 DIV (scRNA-seq). Mean ± SEM from 3 organoids per condition (1 clone × 3 lines).

(B and C) Distribution of ARX$^+$ cells across different cell types (ARX$^+$ cells expressing X marker/total ARX$^+$ cells) in COs (B) and GEOs (C) at 30 and 120 DIV (scRNA-seq).

(D and E) Control COs (D) and GEOs (E) immunostained for ARX, KI67, DCX, and TUJ1 and counterstained with DAPI.

(F and H) COs (F) and GEOs (H) derived from control and ARX^{PAE} hiPSCs, immunostained for ARX and counterstained with DAPI.

(G and I) Percentage of ARX$^+$ area in COs (G) and GEOs (I).

Scale bars: 50 μm. IPCs, intermediate progenitor cells; NPC, neural progenitor cell; oRGC, outer radial glial cell; PC, progenitor cells; RGC, radial glial cell. One-way ANOVA, two-tailed Student's *t* test or non-parametric test; $*p < 0.05$, $**p < 0.01$, $****p < 0.0001$. Mean ± SEM from 14–18 organoids per condition (2 clones × 3 lines). Each point represents an individual organoid. Symbols indicate clones/lines (line 1: clone A •, clone B ○; line 2: clone A ■, clone B □; line 3: clone A ▲, clone B Δ).

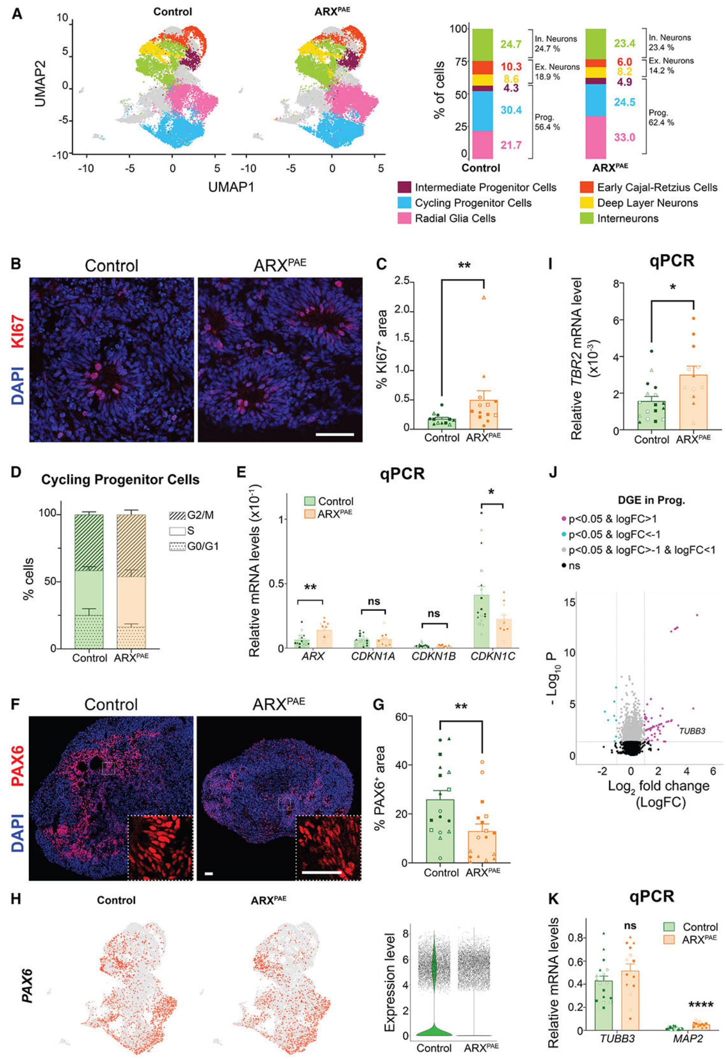

Figure 3. PAE affects progenitors in COs

(A) UMAP plots of control and ARX^{PAE} COs at 30 DIV and a bar graph showing cell cluster percentages.

(B) Control and ARX^{PAE} COs at 30 DIV immunostained for KI67 and counterstained with DAPI.

(C) Percentage of KI67^{+} area in COs at 30 DIV.

(D) Distribution of cycling progenitor cells across cell cycle phases by scRNA-seq.

(E) Relative *ARX*, *CDKN1A*, *CDKN1B*, and *CDKN1C* mRNA levels in COs at 30 DIV by RT-qPCR.

(F) Control and ARX^{PAE} COs at 30 DIV immunostained for PAX6 and counterstained with DAPI.

(G) PAX6^{+} area in control and ARX^{PAE} COs at 30 DIV.

(H) Feature and violin plots showing *PAX6* expression in control and ARX^{PAE} 30 DIV COs.

(I) Relative *TBR2* mRNA levels in control and ARX^{PAE} COs at 30 DIV by RT-qPCR.

(J) Volcano plot of differentially expressed genes (DEGs) in cortical progenitor cells.

(K) Relative *TUBB3* and *MAP2* mRNA levels in COs at 30 DIV by RT-qPCR.

Scale bars: 50 μm. Two-tailed Student's *t* test or non-parametric test; $*p < 0.05$, $**p < 0.01$, $****p < 0.001$; ns, not significant. Mean ± SEM from 12–18 organoids per condition (2 clones × 3 lines). Each point represents an individual organoid (line 1: clone A •, clone B ○; line 2: clone A ■, clone B □; line 3: clone A ▲, clone B Δ).

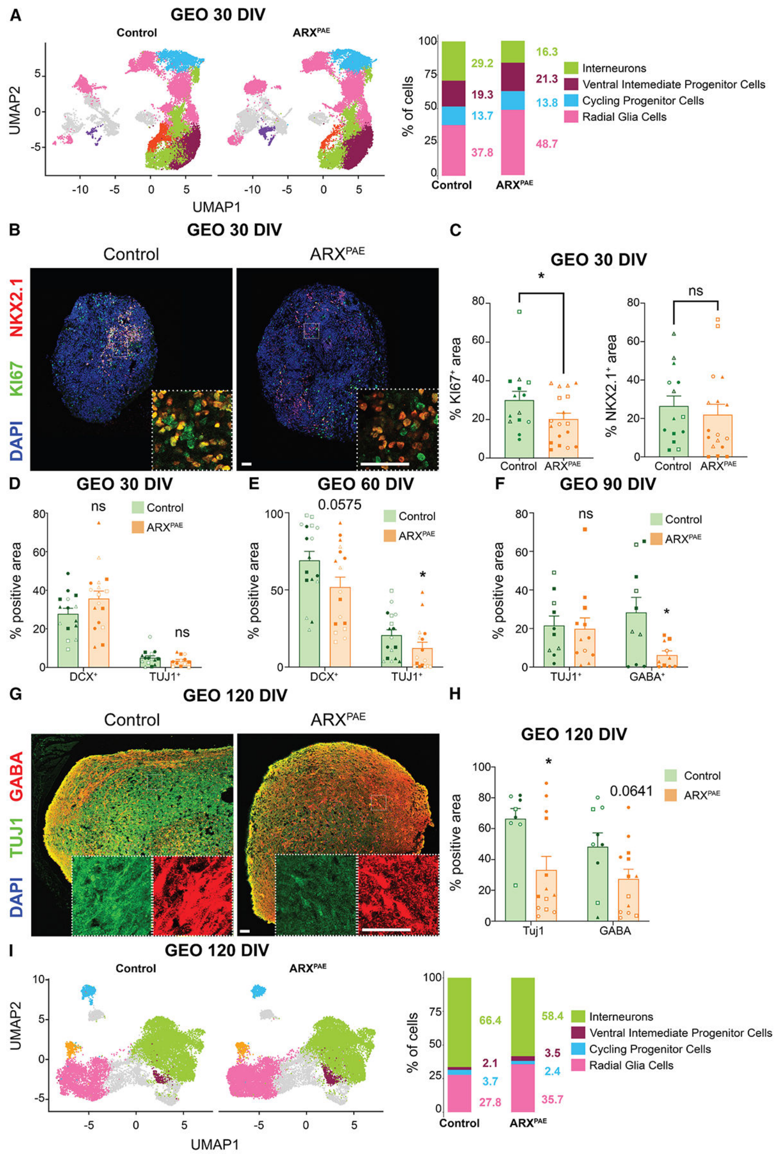

Figure 4. PAE disrupts GABAergic differentiation in GEOs

(A) UMAP plots of control and ARX^{PAE} GEOs at 30 DIV and bar graphs showing cell cluster percentages.

(B) Control and ARX^{PAE} GEOs at 30 DIV immunostained for KI67 and NKX2.1 and counterstained with DAPI.

(C) Quantification of KI67^{+} and NKX2.1^{+} areas at 30 DIV.

(D) Quantification of DCX^{+} and TUJ1^{+} areas at 30 DIV.

(E) Quantification of DCX^{+} and TUJ1^{+} areas at 60 DIV.

(F) Percentage of TUJ1^{+} and GABA^{+} areas at 90 DIV.

(G) Control and ARX^{PAE} GEOs at 120 DIV immunostained for TUJ1 and GABA and counterstained with DAPI.

(H) Quantification of TUJ1^{+} and GABA^{+} areas at 120 DIV.

(I) UMAP plots at 120 DIV and bar graph showing cell cluster percentages.

Scale bars: 50 µm. Two-tailed Student's *t* test or non-parametric test, $*p < 0.05$, ns, not significant. Mean ± SEM from 9–18 organoids per condition (2 clones × 3 lines). Each point represents an individual organoid (line 1: clone A •, clone B ○; clone A ■, clone B □; line 3: clone A ▲, clone B Δ).

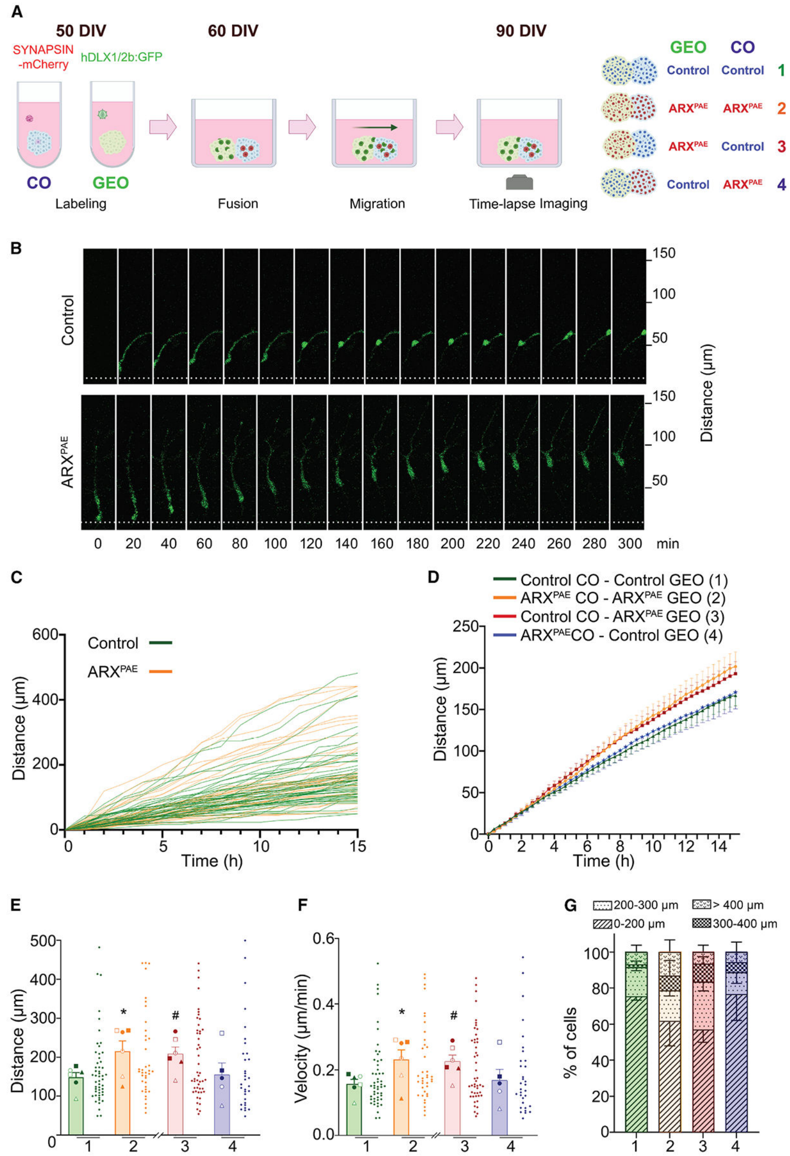

Figure 5. PAE in ARX accelerates interneuron migration in assembloids

(A) Experimental design. GEOs were labeled at 50 DIV with a lenti-DLX1/2b:GFP virus and COs with an AAV-SYNAPSIN-mCherry virus. After 10 days, CO and GEO were fused, and migration was recorded over 15 h at 90 DIV. Four conditions were analyzed: (1) control CO + control GEO, (2) ARX^{PAE} CO + ARX^{PAE} GEO, (3) control CO + ARX^{PAE} GEO, and (4) ARX^{PAE} CO + control GEO.

(B) Time-lapse images of DLX1/2-GFP^{+} interneurons migrating in control and ARX^{PAE} assembloids.

(C) Single-cell migration tracks of DLX1/2-GFP^{+} interneurons over time.

(D) Mean migration distance of DLX1/2-GFP^{+} interneurons over time across conditions.

(E and F) Quantification of total migration (E) and velocity (F) of DLX1/2-GFP^{+} interneurons.

(G) Distribution of migration distances of DLX1/2-GFP^{+} interneurons across conditions.

Mean ± SEM per line; $n = 5–6$ from 32–51 neurons derived from 11–17 organoids (2 clones × 3 lines per condition). Individual neuron values are represented on the right of each bar for each condition. Two-tailed Student's *t* test, *(1 vs. 2) or #(1 vs. 3) $p < 0.05$. Each point represents an individual line. Symbols indicate clone/line (line 1: clone A •, clone B ○; clone A ■, clone B □; line 3: clone A ▲, clone B Δ).

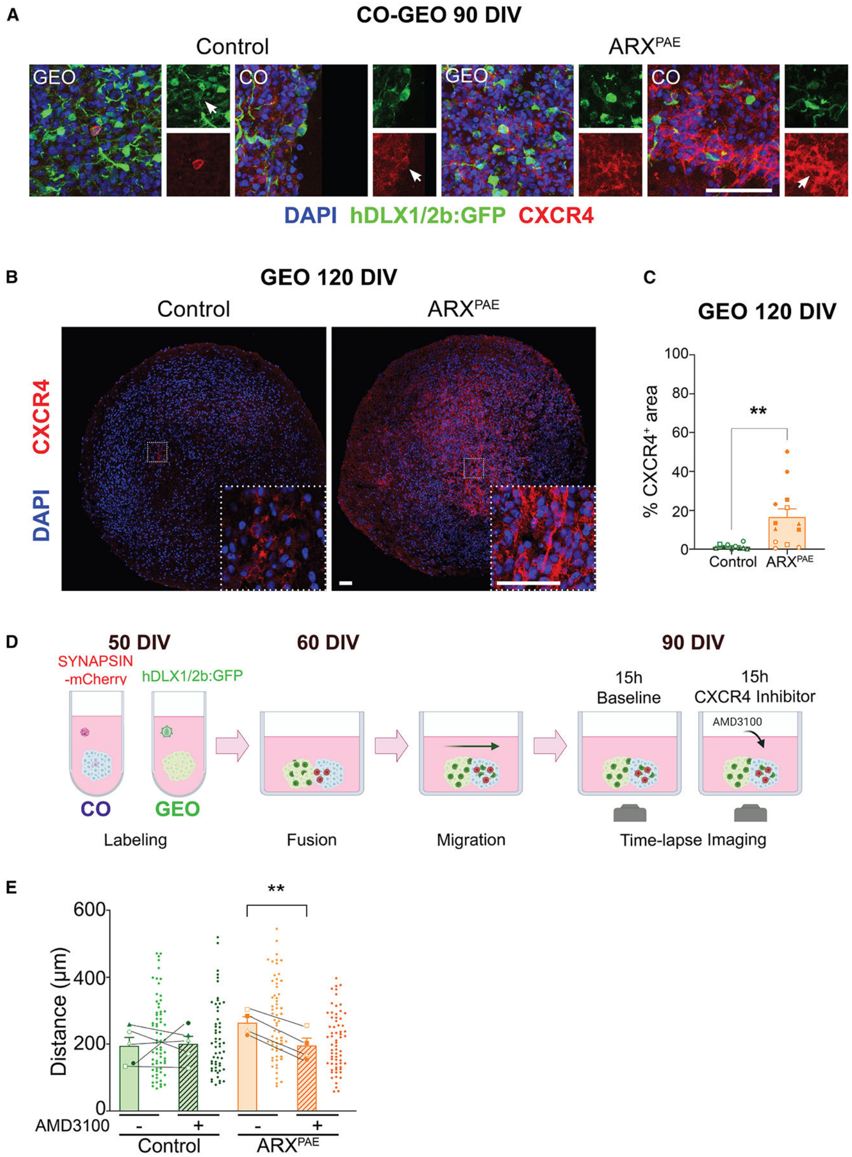

Figure 6. CXCR4 inhibition reduces interneuron migration in ARX^{PAE} assembloids

(A) Control and ARX^{PAE} assembloids at 90 DIV immunostained for GFP and CXCR4 and counterstained with DAPI.

(B) Control and ARX^{PAE} GEOs at 120 DIV immunostained for CXCR4 and counterstained with DAPI.

(C) Quantification of CXCR4^{+} area in control and ARX^{PAE} GEOs at 120 DIV. Mean ± SEM from 13–14 organoids per condition (2 clones × 3 lines). Two-tailed Student's *t* test, $^{**}p<0.01$.

(D) Experimental design. COs and GEOs were fused after labeling, and interneuron migration was recorded for 15 h at baseline and 15 h post AMD3100 (CXCR4 inhibitor) application.

(E) Migration distance of DLX1/2-GFP^{+} cells in control and ARX^{PAE} assembloids before and after AMD3100 treatment.

Data are shown as mean ± SEM per line. $n = 4–5$ from 51–67 neurons derived from 23–24 organoids (2 clones × 3 lines per condition). Individual neuron values are represented to the right of each bar for each condition. Paired t test, $^{**}p < 0.01$. Scale bar: 50 μm. Symbols indicate clone/line (line 1: clone A •, clone B ○; clone A ■, clone B □; line 3: clone A ▲, clone B Δ).

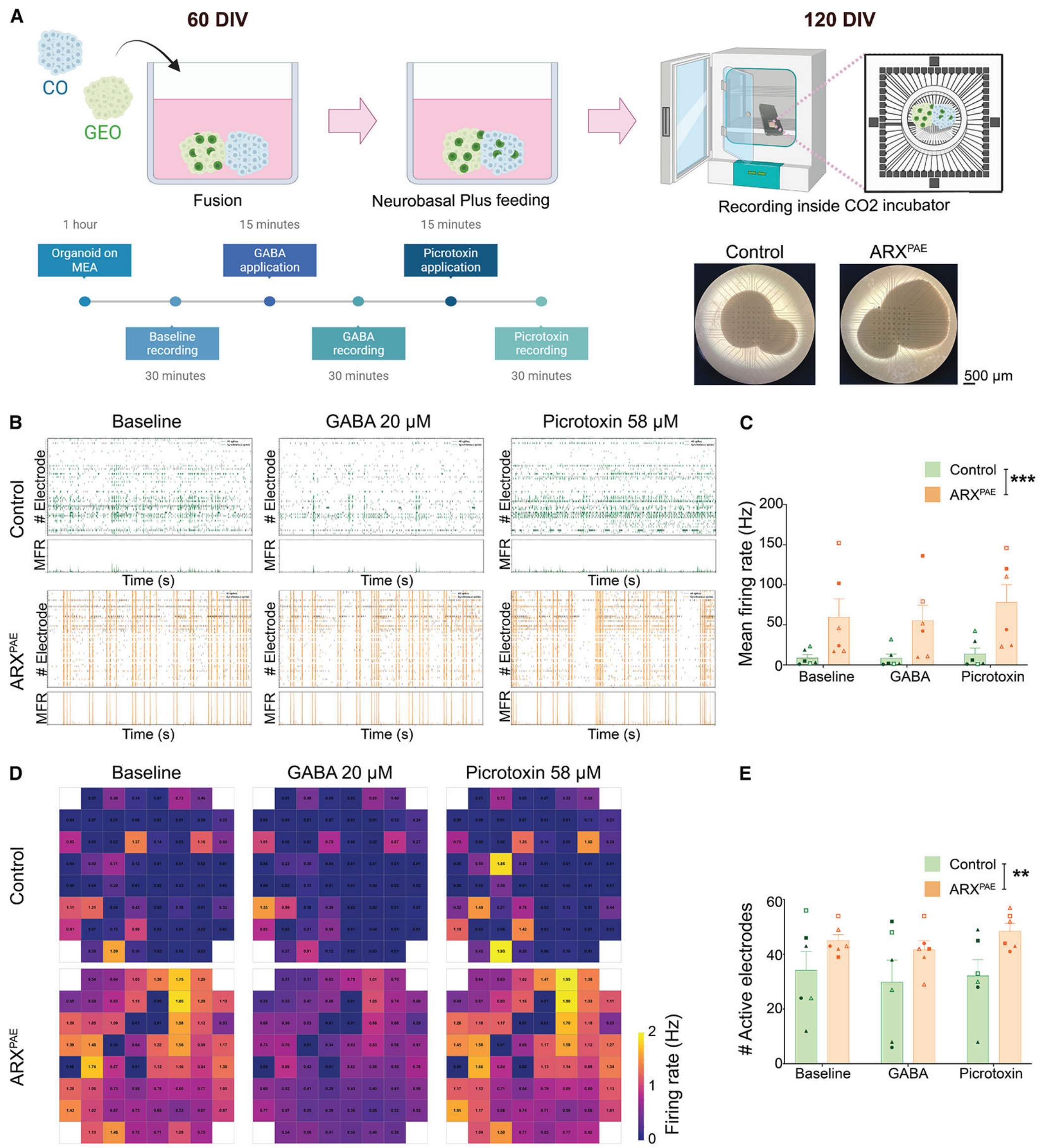

Figure 7. PAE promotes hyperactivity in ARX^{PAE} assembloids

(A) MEA setup. Assembloids on 3D-MEAs were recorded after 1-h rest, baseline, then GABA and picrotoxin application.

(B) 5-min raster plots for control and ARX^{PAE} assembloids at 120 DIV under all conditions; synchronous spikes are color-coded (green, control; orange, ARX^{PAE}).

(C) Mean firing rate (MFR) per condition in 30-minute recordings.

(D) Heatmaps of electrode activity.

(E) Number of active electrodes.

Data are shown as mean ± SEM from 6 organoids from 1–2 clones × 3 lines per condition. two-way ANOVA; $^{**}p < 0.01$, $^{***}p < 0.001$. Each point represents an individual organoid. (line 1: clone A •, Clone B ○; clone A ■, clone B □; line 3: clone A ▲, clone B Δ).

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rabbit Anti-AC3	Cell Signaling	Cat. No. 9661; RRID: AB_2341188
Rabbit Anti-ARX	Gift from Drs. Morohashi and Kitamura	N/A
Mouse Anti-CB	Swant	Cat. No. CB300; RRID: AB_3542811
Mouse Anti-CR	Swant	Cat. No. CR6B3; RRID: AB_10000320
Rat Anti-CTIP2	Abcam	Cat. No. AB18465; RRID: AB_2064130
Rabbit Anti-CXCR4	Invitrogen	Cat. No. PA3-305; RRID: AB_2091817
Guinea pig Anti-DCX	Millipore	Cat. No. AB2253; RRID: AB_1586992
Rabbit Anti-FAM107A	Sigma	Cat. No. HPA055888; RRID: AB_2682957
Rabbit Anti-GABA	Sigma	Cat. No. A2052; RRID: AB_477652
Chicken Anti-GFP	AvesLab	Cat. No. GFP-1020; RRID: AB_10000240
Rabbit Anti-HOPX	Sigma	Cat. No. HPA030180; RRID: AB_10603770
Mouse Anti-Ki67	BD	Cat. No. 550609; RRID: AB_393778
Rabbit Anti-LIN28	Cell Signaling	Cat. No. 3978; RRID: AB_2297060
Mouse Anti-NANOG	Thermoscientific	Cat. No. MA1-017; RRID: AB_2536677
Mouse Anti-NESTIN	Millipore	Cat. No. MAB353; RRID: AB_94911
Rabbit Anti-NKX2-1	Abcam	Cat. No. ab133737; RRID: AB_2811263
Goat Anti-NEUROD1	Santa Cruz	Cat. No. sc-1084; RRID: AB_630922
Mouse Anti-mCherry	Invitrogen	Cat. No. M11217; RRID: AB_2536611
Mouse Anti-Oct3/4	Santa Cruz	Cat. No. sc-5279; RRID: AB_628051
Rabbit Anti-PAX6	Biolegend	Cat. No. 901301; RRID: AB_2565003
Mouse Anti-PV	Millipore	Cat. No. MAB1572; RRID: AB_2174013
Rabbit Anti-SATB2	Abcam	Cat. No. AB34735; RRID: AB_2301417
Rabbit Anti-SOX2	Millipore	Cat. No. AB5603; RRID: AB_2286686
Rat Anti-SST	Millipore	Cat. No. MAB354; RRID: AB_2255365
Rabbit Anti-TBR1	Abcam	Cat. No. AB31940; RRID: AB_2200219
Mouse Anti-TUJ1	Sigma	Cat. No. T8660; RRID: AB_477590
Chicken Anti-Vimentin	Abcam	Cat. No. Ab24525; RRID: AB_778824
Bacterial and virus strains		
hDlx1/2b:GFP lentivirus	Gift from Dr. John Rubenstein	N/A
AAV-DJ1-hSYN1:mCherry	Stanford Gene Vector and Virus Core	N/A
Chemicals, peptides, and recombinant proteins		
mTeSR™1 medium	Stemcell Technologies	Cat. No. 05851
Growth factor-reduced Matrigel	BD Biosciences	Cat. No. 356230
ROCK inhibitor	Selleck Chemicals	Cat. No. S-1049
Versene solution	ThermoFisher	Cat. No. 15040-066
Accutase	Sigma	Cat. No. A6964
DMEM/F12	ThermoFisher	Cat. No. 11330032
Neurobasal A	ThermoFisher	Cat. No. 10888022
B-27 without Vitamin A	ThermoFisher	Cat. No. 12587010

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Knock-out serum replacement	Life Technologies	Cat. No. 10828-028
Glutamax	Gibco	Cat. No. 35050-061
MEM-NEAA	Gibco	Cat. No. 11-140-050
2-mercaptoethanol	Gibco	Cat. No. 21985023
Penicillin/streptomycin	ThermoFisher	Cat. No. 15-140-122
Dorsomorphin	Sigma	Cat. No. P5499
SB-431542	TOCRIS	Cat. No. 1614
bFGF	Peprotech	Cat. No. AF-100-18B
EGF	Peprotech	Cat. No. AF-100-15
BDNF	Peprotech	Cat. No. AF-450-02
NT-3	Peprotech	Cat. No. AF-450-03
IWP-2	Selleckchem	Cat. No. S7085
SAG	Selleckchem	Cat. No. S7779
AMD3100	TOCRIS	Cat. No. 3299
4′,6-diamidino-2-phenylindole	Sigma	Cat. No. D9542
Polyvinyl alcohol solution (PVA)	Sigma	Cat. No. BP168-122
BrainPhys™ hPSC Neuron kit	STEMCELL Technologies	Cat. No. 05795
GABA	Sigma	Cat. No. 56-12-2
Picrotoxin	Sigma	Cat. No. 124-87-8
Critical commercial assays		
Episomal reprogramming kit	Invitrogen	Cat. No. A15960
CytoTune-iPS Sendai Reprogramming kit	Thermofisher	Cat. No. A16517
DNeasy® Blood & Tissue Kit	QIAprep	Cat. No. 69504
OneTaq® Hot Start DNA Polymerase	New England BioLab	Cat. No. M0481
SuperScriptIII First-Strand Synthesis System	Invitrogen	Cat. No. 18080-051
PowerUp™ SYBR™ Green Master Mix	Applied Biosystems	Cat. No. A25741
RNeasy Micro Kit	Qiagen	Cat. No. 74004
Worthington papain dissociation kit	Worthington	Cat. No. LK003150
10x Genomics 3′ Gene Expression Reagent kit v3	10x Genomics	Cat. No. 1000269
Deposited data		
Single-cell RNA sequencing data	This study	NCBI GEO: GSE215362
Experimental models: Cell lines		
hiPSCs Control 1	This study	N/A
hiPSCs Control 2	This study	N/A
hiPSCs Control 3	This study	N/A
hiPSCs Patient 1	This study	N/A
hiPSCs Patient 2	This study	N/A
hiPSCs Patient 3	This study	N/A
Oligonucleotides		
GTTCCGACACCCAGCTTTCATCAG	This study	*ARX* Forward qPCR
TGCCACTGCGCCCTCCAC	This study	*ARX* Reverse qPCR

REAGENT or RESOURCE	SOURCE	IDENTIFIER
CAGGGGAGCAGGCTGAAGGGT	This study	*CDKN1A* Forward qPCR
AGAAGATCAGCCGGCGTTTGGA	This study	*CDKN1A* Reverse qPCR
CGGAGCAATGCGCAGGAATAAG	This study	*CDKN1B* Forward qPCR
GGCATTTGGGGAACCGTCTGA	This study	*CDKN1B* Reverse qPCR
CGGCGCGGCGATCAAGAAG	This study	*CDKN1C* Forward qPCR
GACACGGCGCGGGGACATC	This study	*CDKN1C* Reverse qPCR
GGAAGCTTGTCATCAATGGAAATC	This study	*GAPDH* Forward qPCR
TCAGCAGAGGGGGCAGAGAT	This study	*GAPDH* Reverse qPCR
TTCAAAGAAAAGGCCCAAGCTAAAGTT	This study	*MAP2* Forward qPCR
GCAGGTTGATGCTTCCAGACGAG	This study	*MAP2* Reverse qPCR
AAATGGGTGACCTGTGGCAAAGC	This study	*TBR2* Forward qPCR
GTCTCATCCAGTGGGAACCAGTATTAGG	This study	*TBR2* Reverse qPCR
CAGGGGCCTTTGGACATCTCTTC	This study	*TUBB3* Forward qPCR
TCCGCCCCCTCCGTGTAGT	This study	*TUBB3* Reverse qPCR
GGGGGCCGCCTCCTTCAG	This study	*ARX* Forward PCR
CCTGGTGAAGACGTCCGGGTAGTG	This study	*ARX* Reverse PCR
Software and algorithms		
GraphPad Prism	GraphPad Prism	N/A
10x Genomics Cell Ranger	10x Genomics Cell Ranger	https://www.10xgenomics.com/support/software/cell-ranger/latest
Seurat	Seurat	https://satijalab.org/seurat/
R package ggplot2	ggplot2	https://ggplot2.tidyverse.org/
Harmony	Harmony	https://cran.r-project.org/web/packages/harmony/index.html
EnhancedVolcano	Volcano Plot	https://bioconductor.org/packages/release/bioc/html/EnhancedVolcano.html
Monocle	Monocle-Pseudotime	https://cole-trapnell-lab.github.io/monocle-release/
ImageJ - FIJI	ImageJ	https://imagej.nih.gov/ij/
Python 3.9	Python	https://www.python.org/
Leica Application Suite X	Leica	https://www.leica-microsystems.com/products/microscope-software/p/leica-las-x-ls/
CellProfiler	CellProfiler	https://cellprofiler.org/
FlowJo v10	FlowJo	https://www.flowjo.com/
Biorender	Biorender	https://www.biorender.com/
Other		
Codes	This study	https://github.com/parulvarma123/ARX_Organoids_scRNAseqhttps://doi.org/10.5281/zenodo.17426624https://github.com/mgiugliano/SpiQ